

5. Project Development

FitBuddy — Implemented modules, workflows, APIs, and development details

Implemented Backend

- FastAPI application initialized in app/main.py.
- Database tables created through SQLAlchemy metadata.
- Routes implemented for browser forms, admin view, deletion, JSON APIs, and health checks.
- CRUD layer handles profile upserts, plan creation, retrieval, updates, listing, and deletion.
- Pydantic schemas enforce valid ranges and enumerated values.

AI Integration

The code separates AI responsibilities into a workout-plan generator, nutrition-tip generator, and feedback-driven update function. The README specifies Gemini Pro for workout generation/update and Gemini Flash for nutrition tips, while model names remain configurable in the service layer. If `GOOGLE_API_KEY` is not configured, the project provides deterministic fallback behavior for local testing.

Main Workflow

```
POST /generate-workout
→ validate UserInput
→ save_user()
→ generate_workout_gemini()
→ generate_nutrition_tip_with_flash()
→ save_plan()
→ render result.html

POST /submit-feedback
→ validate FeedbackRequest
→ get_user() + get_latest_plan()
→ update_workout_plan()
→ update_plan()
→ render updated result.html
```

API Surface

- `POST /api/generate-workout` — creates/stores a user plan and returns workout + nutrition data.
- `POST /api/submit-feedback` — revises the latest plan using feedback.
- `GET /api/users` — returns stored user profiles.
- `GET /health` — returns application status and Gemini configuration state.
- `GET /docs` — FastAPI-generated interactive API documentation.

Frontend Implementation

The interface uses three Jinja2 templates. `index.html` collects the profile, `result.html` presents generated content and feedback controls, and `all_users.html` provides the admin/coach view. The

CSS defines a responsive dark interface with cards, form controls, result sections, and mobile breakpoints.

Project basis: This document is derived from the supplied FitBuddy source package, including its README, FastAPI routes, SQLAlchemy models, Jinja2 templates, CSS, and tests.